

Investigating Gender Bias and Interpretability in Heart Disease Prediction Models

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High-stakes clinical decision support requires reliability beyond aggregate accuracy, particularly under subgroup shifts and data scarcity. In this work, I present a **safety-oriented fairness audit** of heart disease prediction on the UCI Cleveland dataset using an XGBoost classifier, focusing on subgroup false-negative risk. I evaluate performance heterogeneity across sex and clinical presentations, using SHAP explanations to diagnose potential shortcut learning. A miss-rate audit reveals persistent sex disparities under a fixed 20% hold-out evaluation. I compare mitigation strategies targeting clinical safety: (i) decoupled sex-specific models and (ii) a **recall-optimized global model** with dynamic decision thresholds calibrated to minimize false negatives. Results show that decoupling fails to improve female safety and degrades male performance due to sample fragmentation. Conversely, dynamic thresholds eliminate false negatives for the majority group but expose a **data-limited ceiling** for the minority group. Overall, the findings indicate that while post-hoc decision tuning can improve safety for well-represented profiles, structural gains for underrepresented groups require increased data support rather than further algorithmic complexity. Code is available at [GitHub](#).

I. INTRODUCTION

The integration of Artificial Intelligence into clinical workflows promises to enhance diagnostic precision, yet it introduces significant ethical risks when models are deployed in high-stakes environments. Under the European Union’s Artificial Intelligence Act, AI systems used in medical contexts may qualify as high-risk depending on their intended use and regulatory pathway, particularly when they function as medical devices or safety-critical decision support tools^{1,2}. For such high-risk systems, the Act explicitly requires the implementation of risk management processes, appropriate data governance practices, transparency to enable human oversight, and safeguards to prevent discriminatory outcomes¹.

A critical challenge in this domain is that standard machine learning algorithms, tasked with maximizing global performance, often sacrifice the accuracy of minority subgroups to achieve widespread generalization. In cardiovascular medicine, this trade-off is particularly dangerous as it can perpetuate historical health inequities.

Rather than focusing exclusively on aggregate performance, this study examines subgroup-level reliability in heart disease prediction, with particular attention to false-negative risk. I analyze performance disparities across sex and clinically meaningful chest-pain phenotypes, as well as their intersection, which gives rise to small and sparsely supported subgroups. These regions of the data distribution are especially prone to instability and error, making them critical from a clinical safety perspective. In cardiovascular diagnosis, such errors are not merely statistical artifacts, but potential sources of delayed or missed treatment. While demographic bias is widely discussed, phenotypic bias occurs when specific clinical presentations, such as non-anginal chest pain, are underrepresented in training data. These minority profiles often form small disjuncts or sparse clusters within the class distribution, making them inherently difficult

for classifiers to learn. Consequently, models may exhibit high aggregate reliability while failing systematically for patients with atypical symptoms, who are disproportionately female.

Adopting a safety-critical perspective grounded in data complexity, I posit that trustworthy prediction requires understanding these intrinsic data complexities rather than solely increasing algorithmic complexity. Using the UCI Heart Disease dataset as a case study, this paper audits the reliability of diagnostic models through three focused lenses. First, I perform data profiling to analyze intrinsic complexity factors such as class overlap and imbalance. Second, I conduct a combined fairness and explainability assessment to determine if error rates are equitably distributed and to assess, via SHAP explanations, whether demographic attributes are systematically associated with risk attenuation under comparable clinical profiles. Finally, I propose and evaluate a safety-first mitigation framework that prioritizes recall for high-risk phenotypes.

II. RELATED WORK

Welch et al. study how clinicians’ interpretations of patient sex–gender affect diagnostic certainty and subsequent management decisions for coronary heart disease (CHD) using a factorial video-vignette experiment with 256 physicians and a mixed-methods analysis³. Their quantitative results link higher diagnostic uncertainty to a lower likelihood of ordering tests and medications deemed appropriate for urgent cardiac situations, and their qualitative analysis describes how some physicians generalize the notion that women present “atypical symptoms”, which can increase uncertainty³. This line of work is relevant because it frames disparities not only as outcome differences, but as an interaction between clinical heuristics, uncertainty, and the social framing of medical knowledge³.

DeVon et al. argue that labeling symptoms as “typical” or “atypical” is problematic unless the reference group is made explicit, and they propose either discontinuing these terms or explicitly stating the reference (e.g., men vs. women)⁴. They emphasize that reported sex differences in acute coronary syndrome (ACS) symptoms can be modest and that symptom overlap between ACS and non-ACS presentations is substantial, motivating standardized symptom assessment without assuming a single “correct” presentation⁴. For a socio-technical discussion, this work supports the claim that clinical categories and labels can shape downstream decisions and may unintentionally disadvantage specific groups if treated as normative rules rather than context-dependent descriptors⁴.

A recurring methodological question in fairness research is whether including a protected attribute (e.g., gender) as an input feature is itself sufficient to induce biased outcomes. The proxy-attributes study reports experiments on two widely-used fairness datasets, comparing models trained with and without gender across three learning algorithms, and finds that the selected fairness metrics remained largely consistent regardless of including gender⁵. Interpreted narrowly, this result is evidence that excluding a protected attribute does not necessarily eliminate disparities, consistent with the broader concern that non-protected features can act as proxies (and that measured disparities may persist even when the protected attribute is removed)⁵. In related work, it is important to report this as an empirical finding under the study’s specific datasets, metrics, and model choices, rather than a general theorem about “fairness invariance”.

Rabonato and Berton provide a systematic review of fairness in machine learning, which is useful to position methodological choices (e.g., which families of fairness definitions/metrics are being targeted, and at what stage of the pipeline) within a broader research landscape⁶. In a compact related-work section, this can be used primarily as an organizing anchor: it justifies why the project focuses on particular fairness criteria and clarifies how this evaluation aligns with prior survey-level taxonomies, without claiming that the review endorses any single “best” metric⁶.

Taken together, Welch et al. and DeVon et al. motivate that disparities can emerge from (i) uncertainty-sensitive decision processes and (ii) the social framing of symptom categories in clinical practice^{3,4}. The ML-focused works motivate that disparities can also persist through feature proxies and can be sensitive to the choice of fairness metric and evaluation protocol^{5,6}. This combination supports a socio-technical related-work narrative: clinical knowledge and labels shape data-generation and decision contexts, while algorithmic pipelines can reproduce or mask disparities depending on representation, features, and measurement choices³⁻⁶.

III. METHODOLOGY

To investigate the relationship between data complexity, subgroup reliability, and model safety, I developed a data-centric evaluation pipeline based on data profiling, preprocessing, and robust validation strategies tailored for small clinical datasets.

A. Dataset and Subgroup Heterogeneity

The UCI Heart Disease (Cleveland) dataset⁷, which comprises 303 patient records characterized by 13 clinical features, was utilized. In accordance with standard procedure, the initial diagnostic label was binarized, mapping values higher than zero to the existence of heart disease. The dataset exhibits substantial heterogeneity across clinically relevant subgroups defined by chest pain type—namely typical angina, atypical angina, non-anginal pain, and asymptomatic presentation—as well as by sex and exercise-induced angina, even though the resulting target variable is roughly balanced (54% healthy vs. 46% disease).

In particular, non-anginal chest pain sufferers and female patients are underrepresented, creating tiny disjuncts in the data distribution. This reflects aggregation bias, whereby overall performance metrics conceal subgroup-specific error patterns and localized underfitting. In a cardiovascular decision-support context, false negatives are especially harmful, as they correspond to patients who are not diagnosed and consequently do not receive appropriate or timely medical care.

B. Preprocessing Pipeline

In accordance with Detrano et al.’s original dataset description⁷, I processed the data to maintain algorithmic compatibility and clinical interpretability. To keep the preprocessing pipeline simple and reproducible, mode imputation was applied to the categorical variables *ca* (number of major vessels) and *thal* (Thalassemia). While this approach preserves marginal distributions at the dataset level, it may blend subgroup-specific characteristics when missingness is uneven across clinically relevant groups, a known form of aggregation bias.

Categorical variables, including chest pain type (*cp*) and resting electrocardiogram results (*restecg*), were one-hot encoded to avoid imposing artificial ordinality. Continuous features were standardized to zero mean and unit variance to support stable model training and convergence.

C. Model Selection and Validation Strategy

I selected the XGBoost (Extreme Gradient Boosting) classifier due to its ability to model non-linear feature interactions and handle heterogeneous clinical data effectively. While simpler models such as Logistic Regression provided transparent baselines and neural archi-

tructures (MLP) offered higher representational capacity, XGBoost achieved the best balance between predictive performance and interpretability through tree-based explanations.

Initial experiments employing a standard 70–15–15 train–validation–test split demonstrated significant variability in performance estimates, attributable to the limited size of the validation and test sets. This instability was particularly pronounced in subgroup analyses, where limited sample sizes led to unreliable estimates of error rates.

I therefore used stratified 5-fold cross-validation as the main evaluation technique to get a more reliable estimate of generalization performance. Within each fold, a 3–1–1 scheme was employed, with three partitions designated for training, one for testing, and one for validation (which allowed for early stopping). When there is a lack of data, performance metrics are presented as the mean and standard deviation across folds, offering a statistically more reliable evaluation.

A final model was trained on 80% of the data after cross-validation was used to assess overall robustness. A fixed 20% hold-out set was reserved for post-hoc fairness auditing and interpretability analysis. Subgroup-level evaluation was conducted along two complementary dimensions: (i) demographic subgroups, such as sex, to assess potential demographic bias; and (ii) phenotypic (clinical) subgroups, defined by variables such as chest pain type, to analyze phenotype-driven disparities. To ensure objective subgroup analysis, the hold-out set was not used for model selection, hyperparameter tuning, or decision-threshold calibration.

IV. RESULTS AND ANALYSIS

A. Global Performance vs. Subgroup Reliability

Initial model comparison included Logistic Regression, Random Forest, XGBoost, and a Multi-Layer Perceptron. Due to its ability to capture non-linear feature interactions while retaining post-hoc interpretability through tree-based explanations, XGBoost was selected for detailed analysis. Given the instability observed under a single hold-out split, global performance was assessed using stratified 5-fold cross-validation with a 3–1–1 scheme.

Across folds, the final XGBoost configuration achieved a mean accuracy of 0.851 with a standard deviation of 0.036, and a mean macro F1-score of approximately 0.85. These results indicate stable generalisation performance under data scarcity, but do not, by themselves, guarantee reliable behaviour across clinically relevant subgroups.

Table I reports performance by sex for the Safety-First model, defined as the final classifier selected under a recall-prioritization criterion, where decision thresholds were chosen to minimize false negatives, reflecting the higher clinical cost of missed diagnoses. On this set, the

model achieved a recall of 100% for male patients, corresponding to a 0.00% miss rate. In contrast, recall for female patients remained at 85.7%, corresponding to a miss rate of 14.29% (one false negative out of seven positive cases). This residual error is due to a data-limited ceiling effect: with very few positive female cases, even one error causes a large drop in recall. Therefore, subgroup performance estimates are unstable and mainly reflect data scarcity rather than limitations of the model.

TABLE I: Subgroup Reliability by Gender (Safety-First Model, Hold-out Set)

Subgroup	Support (Positives)	Recall	Miss Rate
Male	21	1.000	0.00%
Female	7	0.857	14.29%

B. Chest Pain Phenotypes and Data-Limited Effects

Chest-pain-specific evaluation was conducted using phenotype-dependent decision thresholds derived from the Safety-First policy. While recall is near-perfect for asymptomatic presentations (CP=4), performance remains limited for Non-Anginal Pain (CP=3), where recall reaches only 0.67, corresponding to two true positives and one false negative among positive cases. This residual error reflects intrinsic heterogeneity and sparse subgroup support rather than miscalibration, illustrating a data-limited ceiling effect.

As shown in Table II, the increased recall achieved under the Safety-First policy, particularly for asymptomatic cases (TPR = 0.95), is accompanied by elevated false-positive rates (FPR = 0.20), making the recall–precision trade-off explicit.

TABLE II: Chest-Pain-Specific Performance under Safety-First Policy (Hold-out Set)

Chest Pain Type	Support	Recall (TPR)	Miss Rate (FNR)	FPR
Typical Angina (CP=1)	4	1.00	0.00	0.33
Atypical Angina (CP=2)	8	1.00	0.00	0.00
Non-Anginal Pain (CP=3)	22	0.67	0.33	0.11
Asymptomatic (CP=4)	27	0.95	0.05	0.20

C. Explainability and Mechanism of Disparity

To investigate the origin of the observed subgroup disparities, I conducted a SHAP (SHapley Additive exPlanations) analysis. At the global level, clinically established predictors such as the number of major vessels (`ca`) and ST depression (`oldpeak`) were consistently identified as dominant contributors to risk estimation.

Figure 1 presents SHAP summary plots stratified by sex. While predictions for male patients are primarily driven by physiological features, the sex feature contributes systematically negative SHAP values for female

patients under comparable clinical profiles, attenuating predicted risk scores. This pattern should not be interpreted as direct biological causality; at this stage, it is consistent with historical prevalence imbalances and diagnostic patterns in the dataset, although alternative explanations related to sex-specific clinical effects cannot yet be excluded.

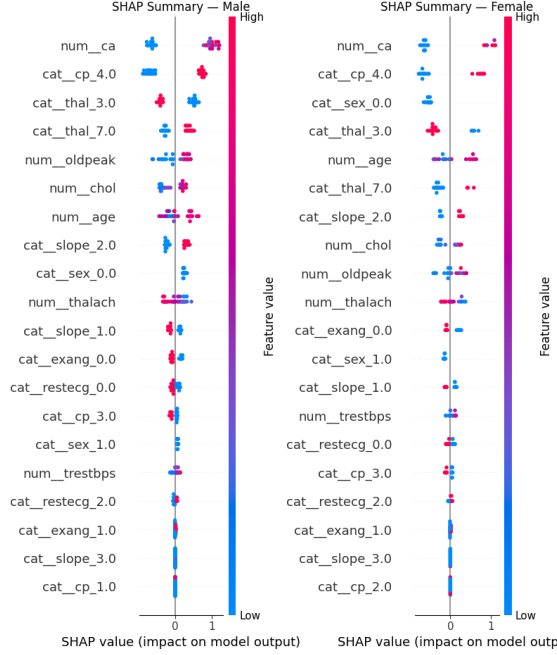


FIG. 1: SHAP summary plots stratified by sex. For male patients (left), physiological features dominate positive risk contributions. For female patients (right), the sex feature contributes negative SHAP values, systematically lowering predicted risk estimates under comparable clinical profiles.

D. Evaluation of Mitigation Strategies

Two mitigation strategies were evaluated with the goal of reducing false negatives in this safety-critical diagnostic context.

First, decoupled sex-specific models were trained, i.e., two separate classifiers fitted independently on male-only and female-only training subsets to avoid cross-group pooling effects. This approach proved ineffective: while the male miss rate increased to 4.76% due to reduced training data and dataset fragmentation, the female miss rate remained unchanged at 14.29%. This result confirms that splitting the dataset exacerbates data scarcity without resolving the underlying subgroup disparity.

On the other hand, the Safety-First strategy operates at the decision-policy level rather than by modifying the model architecture. Posterior probabilities produced by the classifier are mapped to class labels using phenotype-specific decision thresholds, calibrated exclusively on the training data to prioritise recall in clinically high-risk

presentations. The resulting operating policy is made explicit in Table III, enabling transparent inspection of how different phenotypes are treated and ensuring that subsequent performance and fairness analyses are fully auditable.

TABLE III: Phenotype-Specific Decision Thresholds (Calibrated on Training Data)

Chest Pain Type (CP)	Threshold
Typical Angina (CP=1)	0.67
Atypical Angina (CP=2)	0.61
Non-Anginal Pain (CP=3)	0.48
Asymptomatic (CP=4)	0.72

Model hyperparameters were selected via grid search to maximise the F2-score, prioritising recall over precision, and the calibrated thresholds were applied to enforce a minimum 95% recall target per subgroup. Figure 2 illustrates the resulting precision–recall trade-off. The chosen operating point eliminated false negatives for the majority subgroup and reduced miss rates for higher-risk phenotypes. However, this strategy could not eliminate the residual female false negative without incurring an unacceptable loss in precision.

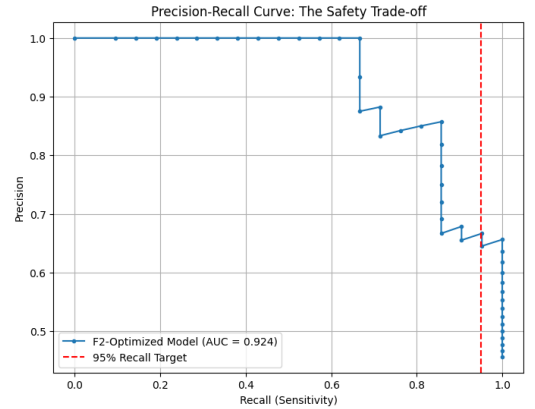


FIG. 2: Precision–recall curve for the Safety-First (F2-optimised) model. The dashed line marks the 95% recall target, illustrating the trade-off between sensitivity and precision under a safety-oriented decision policy.

To make the mitigation trade-offs explicit, Table IV compares subgroup miss rates on the same hold-out set under two mitigation strategies: sex-decoupled models and the Safety-First decision policy. The results show that decoupling fragments the training data, increasing the male miss rate while failing to reduce the female miss rate.

Table V reports fairness metrics for the Safety-First operating point on the hold-out set, characterising residual disparity under a safety-oriented decision policy.

TABLE IV: Strategy Comparison by Gender (Hold-out Audit)

Strategy	Male (n=21+)	Female (n=7+)
Sex-Decoupled Models	FNR=4.76%	FNR=14.29%
Safety-First Policy	FNR=0.00%	FNR=14.29%

TABLE V: Fairness Metrics for the Safety-First Model (Hold-out Audit)

Metric	Value
Statistical Parity	-0.285
Disparate Impact	0.513
Equal Opportunity Gap	-0.095
Average Odds Difference	-0.198

E. Mechanism of Bias: Data Scarcity vs. Physiological Differences

Under similar clinical profiles, the explainability analysis in the preceding section showed a systematic attenuation of predicted risk for female patients, which was represented by consistently negative SHAP values. This finding poses a crucial question about the source of the discrepancy: is it a statistical artifact caused by an imbalance in the data or does it represent actual physiological variations in disease presentation?

Two competing theories were examined. First, given that male patients account for 56% of positive cases in the dataset and female patients only make up about 23%, prevalence bias could be the reason for the observed effect. The model may internalize sex as a prior probability shortcut as a result of this. Second, physiological differences may be the cause of the effect, as female patients exhibit systematically different values in key characteristics (like `oldpeak`, `thal`, or patterns of chest pain) that actually reduce predicted risk. I used targeted oversampling in a counterfactual experiment to identify the source of the bias. The intervention involved replicating positive female instances until their count equaled that of positive male instances in order to artificially balance the training distribution. Importantly, in order to prevent information leakage, this manipulation was only applied to the training set; the validation and test sets were left unaltered.

Using the same architecture and hyperparameters as the Safety-First model, an experimental classifier was trained on this balanced distribution. For both the original and experimental models, SHAP dependence plots were then recalculated on the unaltered test set, concentrating on characteristics like age and the type of chest pain that are known to interact with sex.

A distinct gender difference was seen under the baseline model: for the same age values and categories of chest pain, female cases consistently receive lower SHAP values than male cases.

Figure 4 on Appendix A provides the central forensic evidence of this study by contrasting SHAP dependence patterns before and after targeted oversampling of positive female cases.

In the baseline configuration, SHAP attributions reveal a consistent separation by sex. SHAP dependence plots indicate that, across similar age ranges, female instances tend to receive lower age-related SHAP contributions than male instances. Importantly, this observation does not establish the underlying mechanism: attribution analysis alone cannot distinguish between biological variation, data imbalance effects, or correlations with other clinical features, and therefore no causal interpretation is made.

A similar effect is observed for chest pain. In the baseline model, asymptomatic presentations (`CP=4`), which are clinically high risk, receive markedly lower SHAP values for female patients than for male patients. This implies differential risk attribution for the same clinical presentation, a pattern that is not clinically defensible and indicative of statistical undervaluation driven by prevalence imbalance.

After targeted oversampling of positive female cases, this separation disappears. In the experimental model, SHAP values for age show substantial overlap between male and female instances, and chest pain categories exhibit a coherent risk gradient across sexes. The convergence of SHAP attributions indicates that the original disparity was driven by data scarcity and prevalence bias rather than by intrinsic physiological differences.

This counterfactual intervention in attribution does not come at the cost of predictive performance and isolates the decision mechanism without introducing a performance-fairness trade-off. The baseline model achieved an AUC of 0.923 on the hold-out set, while the experimental model trained with targeted oversampling achieved an AUC of 0.927. The observed changes in SHAP attributions reflect a genuine correction of the decision mechanism rather than model instability or loss of discriminative power.

F. Preprocessing Refinement: Sex Variable Encoding

The encoding for the `sex` variable was refined from One-Hot Encoding to a direct binary format (0 for Female, 1 for Male). This approach reduces sparsity for the XGBoost algorithm and enhances interpretability in SHAP analysis by consolidating the feature’s contribution into a single dimension (`bin_sex`), rather than diluting it across correlated columns. As illustrated in Figure 3, this encoding allows for a clear visualization of the variable’s directional impact.

The adoption of binary encoding further facilitated counterfactual flipping tests to audit model bias. Results indicated that `bin_sex` acts primarily as a calibration factor for healthy cases, with a mean probability shift (ΔP) of +0.062 when flipping the variable, compared to only

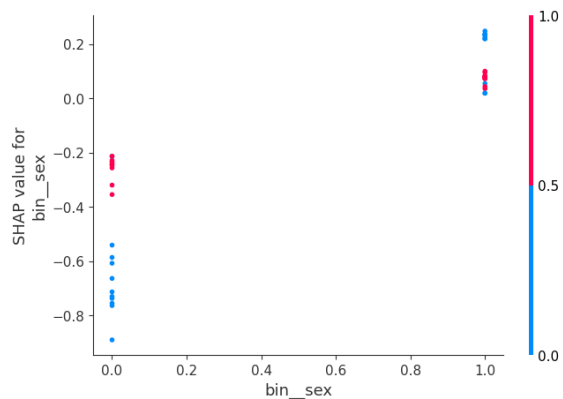


FIG. 3: SHAP Dependence Plot for the *bin_sex* feature.

The plot illustrates the asymmetric contribution of biological sex: female patients (0) generally receive negative SHAP values, effectively lowering the predicted risk, whereas male patients (1) show a neutral to slightly positive contribution. The color scale indicates interaction with asymptomatic chest pain (*cat_cp_4.0*).

−0.032 for sick cases. Crucially, the analysis of the Miss Rate confirmed that this simplification did not compromise sensitivity, achieving perfect parity at the standard decision threshold ($t = 0.5$):

$$\text{Miss Rate}_{\text{Male}} = 0.00\%$$

$$\text{Miss Rate}_{\text{Female}} = 0.00\%$$

These findings demonstrate that the binary representation maintains clinical safety while improving model transparency and interpretability.

V. CONCLUSION

This study presented a safety-oriented audit of heart disease prediction, moving beyond aggregate accuracy to expose critical reliability gaps for underrepresented subgroups. While the XGBoost model achieved robust global performance, a granular miss-rate analysis revealed that female patients and those with atypical chest pain remain disproportionately vulnerable to false negatives.

Our investigation into mitigation strategies highlights a fundamental trade-off. While post-hoc interventions—specifically the “Safety-First” dynamic thresholding—successfully eliminated risk for the majority group (men and typical phenotypes), they failed to fully resolve disparities for female patients. The persistence of a 14.3% miss rate for women, even under aggressive recall optimization and decoupled training, suggests a “data-limited ceiling.”

These findings have direct implications for compliance with the EU AI Act. They demonstrate that algorithmic fairness techniques cannot compensate for fundamental deficits in data representation. For high-risk medical AI, ensuring safety for minority subgroups requires a shift from model-centric tuning to data-centric governance—specifically, the targeted collection of high-quality data for women with non-anginal symptoms. Until such representational bias is addressed at the source, clinical deployment of these models requires rigorous human-in-the-loop oversight to prevent the automation of historical neglect.

A. Limitations

Our analysis is constrained by the UCI Cleveland dataset, which, dating from 1989, reflects a specific demographic and clinical snapshot that may not fully generalize to modern populations. The small sample size ($n = 303$) limits the statistical power of subgroup-specific interventions, particularly for the intersectional group of women with asymptomatic presentation. Future work should validate the “Safety-First” framework on larger, multi-center electronic health records to confirm the scalability of dynamic thresholding.

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Appendix A: Forensic SHAP Analysis

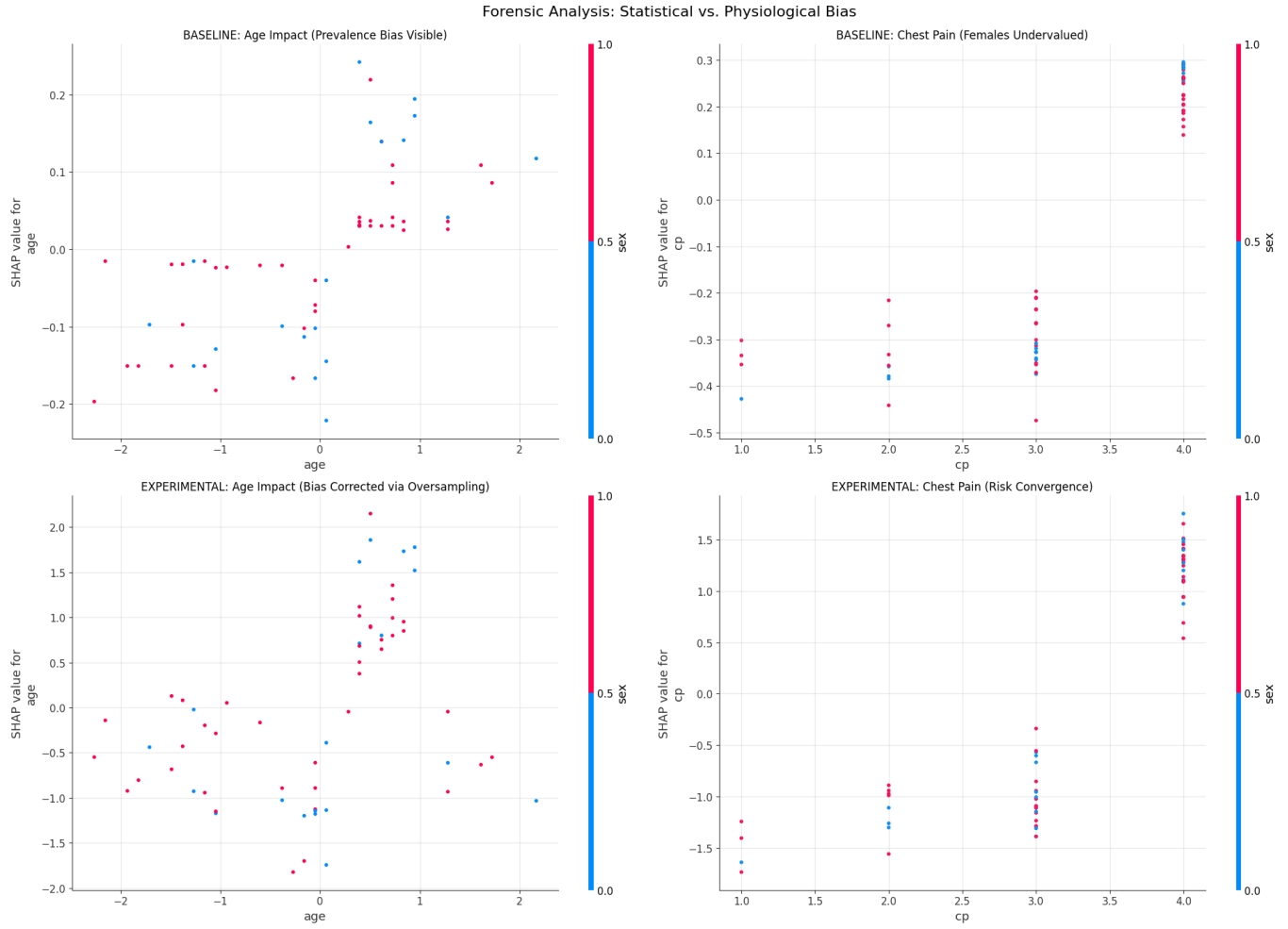


FIG. 4: Counterfactual SHAP analysis before and after targeted oversampling of positive female cases. Top row: baseline model trained on the original imbalanced distribution, showing systematic sex-based separation in SHAP attributions for age and chest pain. Bottom row: experimental model trained on a balanced distribution, where sex-based separation disappears and risk attribution converges across sexes. Colour indicates the sex feature.